Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/14/2005
PI Name:	Kandarian, Susan Ph.D.		
Project Title:	Gene Expression Profiling of Unloaded Skeletal Muscle		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02215	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
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No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Kandarian, Susan		
COI Name (Institution):			
Grant/Contract No.:	MA.002.07		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The overall aim of the proposed work is to delineate cellular mechanisms involved in unloading-induced skeletal muscle atrophy. As a first step in elucidating these mechanisms we have analyzed global gene expression patterns in rat soleus muscles using Affymetrix GeneChips at 1, 4, 7, 14 days of biomechanical unloading. Temporal analysis of the parallel expression of ~8000 gene products has served as a window into the signaling networks that underlie the atrophy process. Microarray data analysis software has been used to segregate gene expression changes based on involvement in known functional groups, regulatory and signaling pathways. Clustering algorithms have been used to elucidate sets of co-regulated genes with similar temporal expression patterns. We are currently in the process of testing candidate genes that may be involved in the progression of atrophy using overexpression systems in both cell culture and in whole muscle.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is well known that exposure to microgravity, as well as muscular unloading on earth, leads to marked decreases in muscle size, functional capacity, and fatigue resistance. What is poorly understood about atrophy is how the protein loss is controlled, including the triggers and signaling mechanisms. Results from the microarray analysis of unloaded rat soleus muscles have not only given us insight into the processes and pathways involved in muscle atrophy but they have provided an excellent source of candidate genes to test for regulatory roles in the atrophy process. In identifying these genes we will be in a better position to develop more effective countermeasures to combat the deleterious changes in muscle function due to exposure to spaceflight. The CRLs addressed in this research are #2-4; we will identify genes involved in muscle atrophy and thereby identify targets for countermeasures, either nutritional or pharmacological.
Task Progress:	Muscular inactivity leads to atrophy, weakness, and decreased fatigue resistance. In order to provide a window into the dynamic processes that underlie muscle atrophy, we performed global gene expression analysis of rat soleus muscles using Affymetrix GeneChips at 1, 4, 7 and 14 days of hindlimb unloading. Expression of 309 known genes was significantly changed by at least 2-fold (212 upregulated, 97 downregulated). K-means clustering was used to divide these genes into co-regulated clusters based on the similarity of temporal expression patterns. This allowed the development of a timeline of the atrophy process with respect to the behavior of genes in a broad array of functional categories. Regulatory genes were often upregulated early, in either a transient or sustained manner, but they also populated clusters with later patterns of activation, suggesting different phases of molecular adaptations. Other early events were the activation of ubiquitination genes and downregulation of protein chaperones. In comparison, clusters representing slightly delayed activation patterns included genes involved in fast contraction, glycolysis, translational inhibition, oxidative stress, protein degradation, and amino acid catabolism. Downregulated genes exhibited fewer unique expression patterns and included structural and regulatory genes of the extracellular matrix and cytoskeleton, and genes that define a slow-oxidative phenotype. Other novel findings include the tight co-activation of proteasome subunit and ubiquitination genes, differential regulation of serine proteases and serine protease inhibitors, and the identification of transcriptional, signaling, growth and cell cycle genes that likely play a role in the atrophy process. The present work has uncovered temporal patterns of gene expression that highlight the molecular processes that underlie muscle atrophy and provide new avenues for study
Bibliography Type:	Description: (Last Updated: 09/05/2007)
Articles in Peer-reviewed Journals	Stevenson, E.J. and S.C. Kandarian "Gene expression profiling shows its muscle. " Physiology News. 54: 24-25 , Feb-2004
Articles in Peer-reviewed Journals	Jackman, R.W. and S.C Kandarian "The molecular basis of skeletal muscle atrophy" Am J Physiol Cell Physiol. 287(4):C834-843 , Oct-2004